

Micro-controller Basics

→ microprocessor ~~है~~ CPU

→ Micro-controller ~~है~~ पूरा computer ~~है~~
 इसमें microprocessor ~~है~~ ~~संस्थापित~~ ~~है~~ processing
 unit ~~है~~ ~~है~~। इसके external ~~है~~ add ~~है~~
~~है~~ ~~है~~ ~~है~~

Microprocessor vs micro-controller

Feature	microprocessor	micro controller
i) Main compon component	only CPU	CPU + RAM + ROM + ...
ii) External parts	प्रत्येक है	Normally है प्रत्येक है है
iii) purpose	General purpose	specific purpose
iv) performance	बुरा	उत्कृष्ट है है
v) cost	बुरा	अच्छा
vi) power consumption	अधिक	अल्प
Example	PC, Laptop	Arduino...

Criteria for choosing a micro-controller

i) 2nd bit for MC hardware based project or software

Arduino nano → 8 bit

Arduino Uno → 16 bit

Arduino mega → 32 bit

ii) highest speed

iii) power consumption

iv) How much RAM and ROM is available on that chip

v) I/O pin cost

vi) Analog → digital conversion support

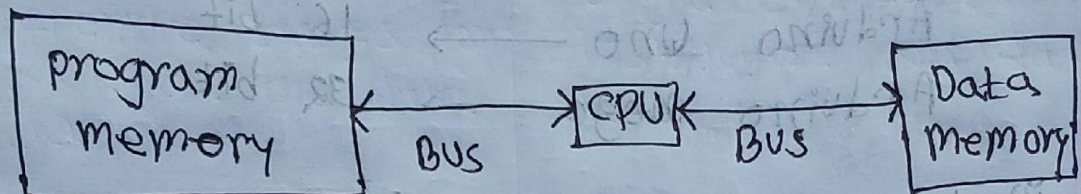
vii) cost per unit

Raspberry pi vs Arduino

Raspberry pi	Arduino
i) MP based single board computer	i) MC based physical computing platform
ii) Runs a full fledged linux OS	ii) Runs a single program only
iii) Doesn't support analog I/O	iii) Support analog I/O
iv) 1.2 GHz processor with 1 GB RAM	iv) 16 MHz processor with 2 KB RAM (Uno)

Types of computer Architecture:

1) Harvard Architecture



→ Fast

→ can work in parallel

→ more memory for program & data individually

→ can handle complex task, But Architecture is also complex

→ Higher cost

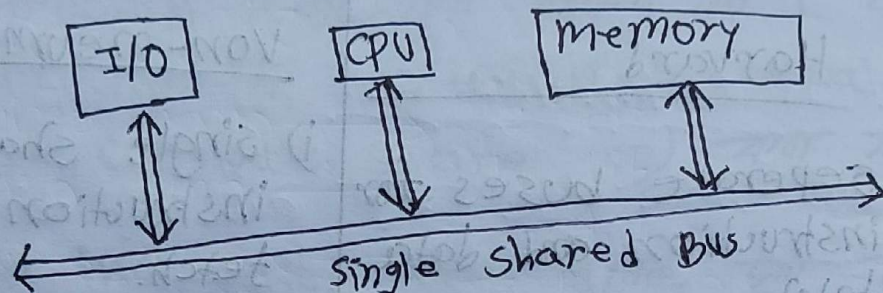
→ Wasted space

Data memory is full but instruction memory is 80% empty, we can't use that 80%

→ memory can be different characteristics (size different etc)

→ Instruction part can be kept read only. But data memory can be both read & write permitted (General role)

ii) von Neuman Architecture



Stores both the computer's instruction (code) & the data in the exact same physical memory space, using a single shared pathway.

- cost efficient
- flexibility in storing data & memory. shared space allows full ~~util~~ utilization
- simple design

✗ no parallelism. Low speed. In Instruction fetch & data operation can't ~~be~~ be done at the same time as they use single shared bus
von Neuman Bottleneck.

✗ computer can accidentally mistake data for actual code. makes it vulnerable to Buffer overflow attack.

Harvard Vs Von-neuman

Harvard	Von-neuman
i) separate buses for instruction and data fetch	i) single shared bus for instruction and data fetch.
ii) High performance	ii) Low performance.
iii) High cost (2 memory)	iii) Low cost (1 memory)

Risc Versus CISC

Risc Reduced instruction set computing

Mainly 80% ke instructions CPU me common hote hain jaise (Load, Add, Store etc) aur bas use ke liye 20% ke instructions ki zaroorat hai. Risc mainly ke common instruction set me hote hain.

→ Har ek instruction same size ki hai aur same "1 clock cycle" time ke andar execute hoti hai.

Load A, 2:3
Load B, 3:5
Store 2:3, A
Prod A, B

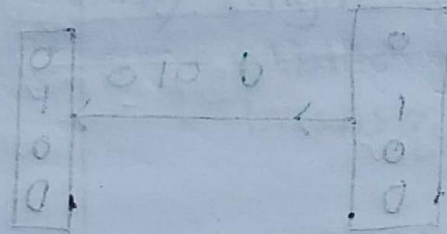
Load A, 2:3
Load B, 3:5
Prod A, B
Store 2:3, A

CISC

complex instruction set computer.

- instruction set का बड़ा दायरा है। यानी 2 टि instruction कि इसके अंदर internal काम (Load, calculate, store ...) कर रहा है।
- इसका instruction cycle time कम होता है।
- Allows arithmetic operation directly into Hardware.

mult: 2:3, 5:2



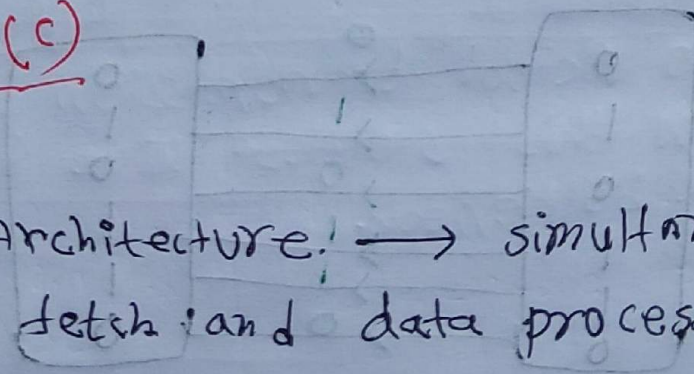
Fall 25 Q1 (C)

• Von-neuman Architecture as cost is low and not much complexity is there.

summer 25 Q2 (C)

Cisc → direct hardware operation करता है।

spring 25 Q2 (C)



Harvard Architecture → simultaneously instruction fetch and data processing.

Communication protocol

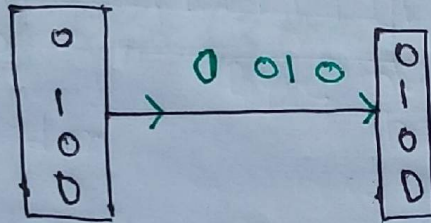
serial communication

একটি Wire Use করে একে একে bit transfer করে।

Ex:

Sender

Receiver



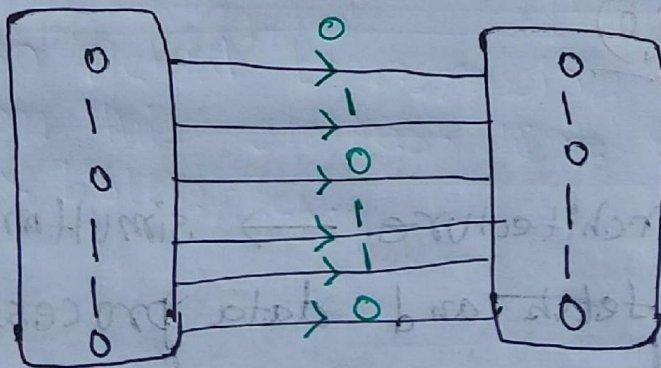
parallel communication

অনেক স্লট Wire Use করে অনেক স্লট bit একসাথে transfer করে।

Ex:

Sender

Receiver



serial vs parallel

serial communication

- i) one data bit is transceived at a time
- ii) slower
- iii) Less number of cables required to transmit data

parallel communication

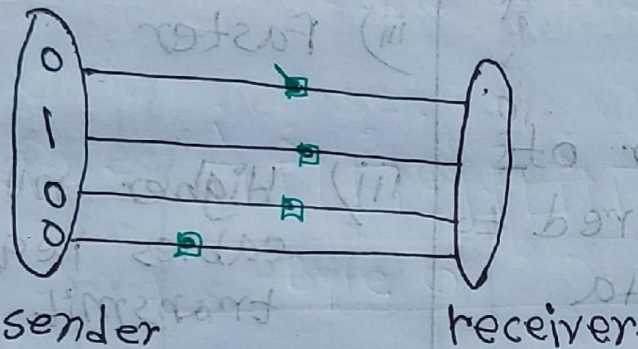
- i) Multiple data bits are transceived at a time
- ii) Faster
- iii) Higher number of cables required to transmit data

Transceived $\rightarrow$ Transmit + received

problem with parallel communication:

i) clock skew

Time difference in the arrival of two bits sequentially in adjacent registers.



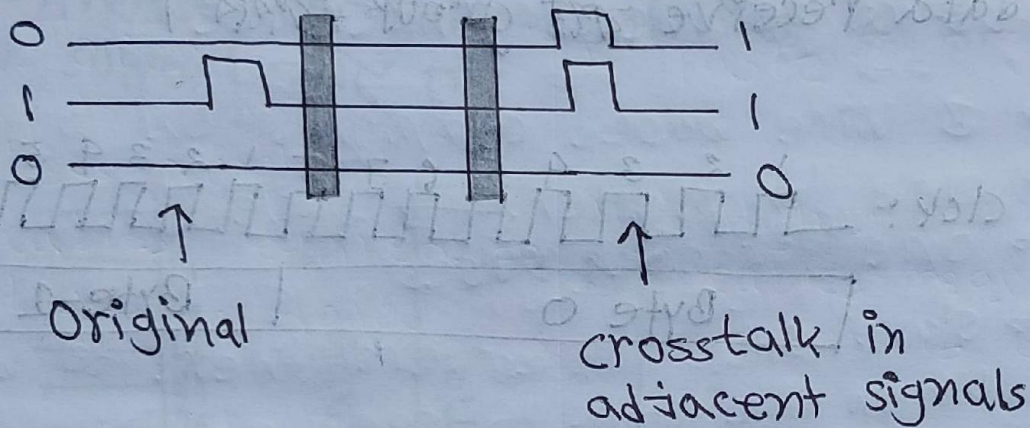
विद्यमान तारों पर 4 वीर wire use कराई गई है। फिर
 normal 2 वीर wire द्वारा transmission rate/
 speed different रहेगा। यह एक सिंगल Bits
 द्वारा प्रोड्यूस सिंगल bit पर प्रोड्यूस प्रोड्यूस।

50 লাইট Bit এর slower bit এর
 দ্বিতীয় round second round transmit করে
 পরে যাবে।

so Actual transmission speed is depend on this slowest wire & it creating a bottleneck.

ii) Crosstalk :

Wire গুলো parallelly মাফার কারণে প্রতি
Wire এর voltage দানের wire এর effect
এর induction (আবেগ) এর কারণে।



বিভিন্ন crosstalk এর কারণে 010 পাঠিয়েছি কিন্তু
receiver পাঠিয়ে 110, আর ~~আসল~~ আসল
কারণে first wire এর voltage change হচ্ছে।

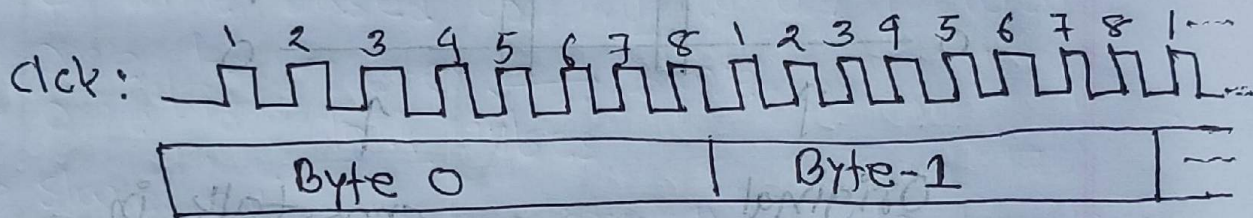
Advantage of serial

- i) No crosstalk
- ii) No clock skew
- iii) Fewer wires required.

Serial communication can be two types:

i) Synchronous:

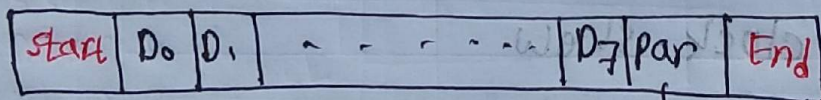
एक ही start or stop bit के साथ data भेजा जाता है, receiver भी उसी time पर data receive कर group करता है।



ii) Asynchronous:

प्रति Byte के लिए start bit (0) और end bit (1) के साथ Receiver को भेजा जाता है।

Due to start and end bit some extra space is required.

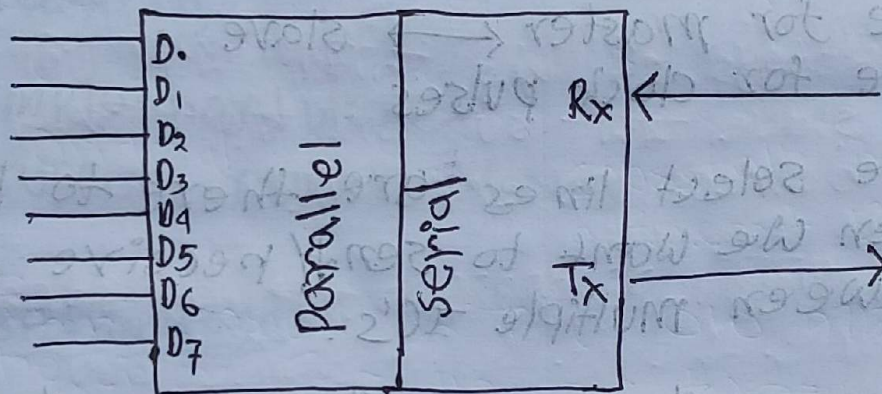


Parity Bit → mainly for error detection.

UART:

Universal Asynchronous Receiver transmitter.

फिर main काम रला **parallel $\longleftrightarrow$ serial** conversion or microcontroller एर र्शि method एर support करे देर



USART:

Universal synchronous Asynchronous Receiver transmitter.

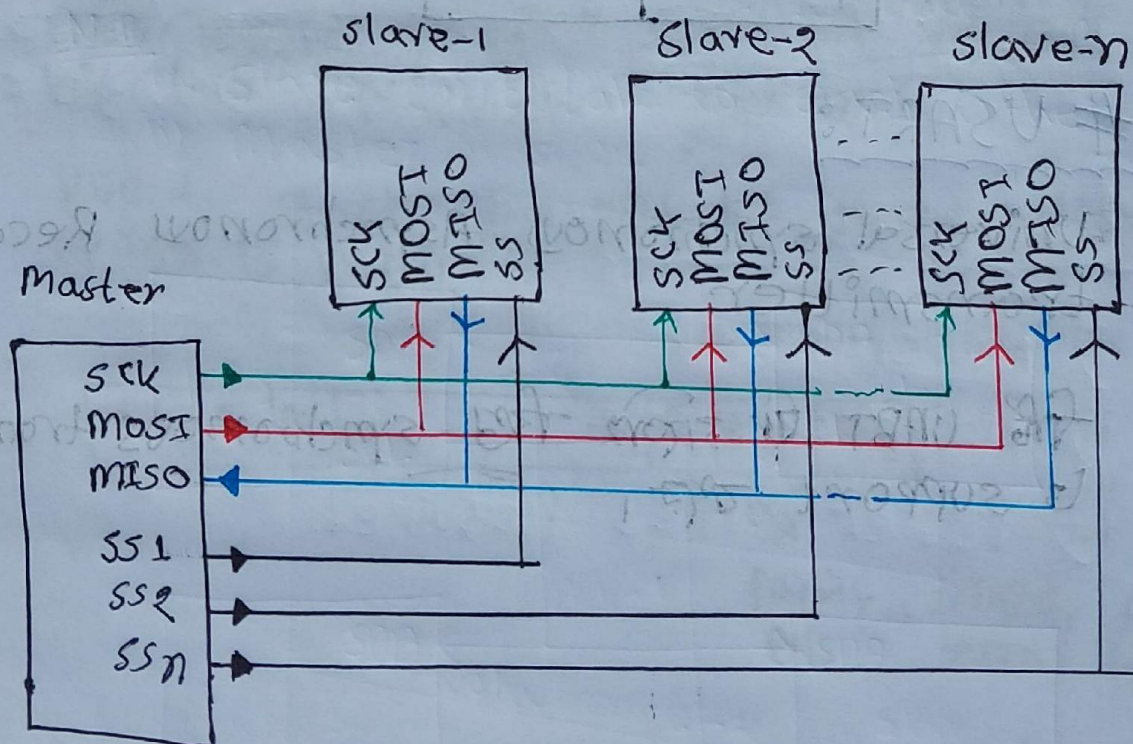
फिर UART एर र्शि सुरु synchronous transmission एर support करे।

serial communication protocols

1) SPI

→ Serial peripheral Interface

- 3 Wire based communication protocol
- One for master ↔ slave
- One for clock pulses
- slave select lines are there to be used when we want to send/receive data between multiple IC's.



MOSI $\rightarrow$ master out, slave in (master $\xrightarrow{\text{data}}$ slave)
MISO $\rightarrow$ master in, slave out (master $\xleftarrow{\text{data}}$ slave)
SCK $\rightarrow$ serial clock
SS $\rightarrow$ slave select (SS=1, selected)

only one master is allowed.

Advantage

- i) simple hardware
- ii) support multiple slaves

Disadvantage

- i) All communication has to be controlled by master slave.

$\rightarrow$ Everything is controlled by clock, and clock is controlled by master.

- ii) Every slave has its own pin slot for SS, MOSI, MISO and SCK.

i) I2C:

inter-integrated circuit (I2C)

$\rightarrow$ Requires only 2 signal wires to exchange information.

→ Allows to have multiple slaves who can communicate with multiple masters

→ can support upto 1008 slaves.

→ 2 length of address

→ 7 bit $\rightarrow 2^7 \rightarrow 128$
→ 10 bit $\rightarrow 2^{10} \rightarrow 1024$

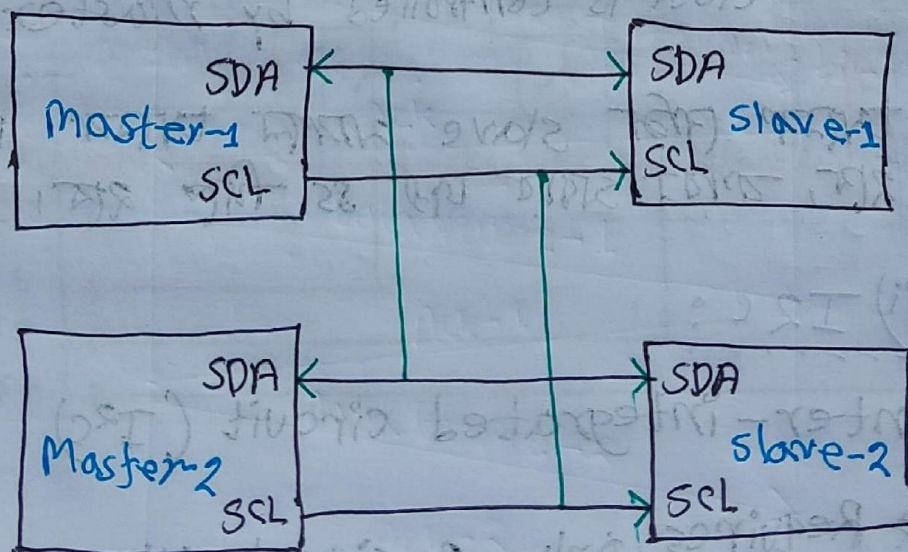
always 2nd bit always zero slave support only, 1st bit address reserve

0000000000 $\rightarrow$ Broadcast

N.B

i) slave can't talk to each other

ii) master " " " "



SCL → serial clock line

SDA → serial Data ~~transfer~~

↳ Bidirectional, can both receive and transmit data

Advantage,

i) Flexibility to use multi-master and multi-slave

ii) simplicity of only two pins

iii) Addressing feature.

iv) Reliable data transfer

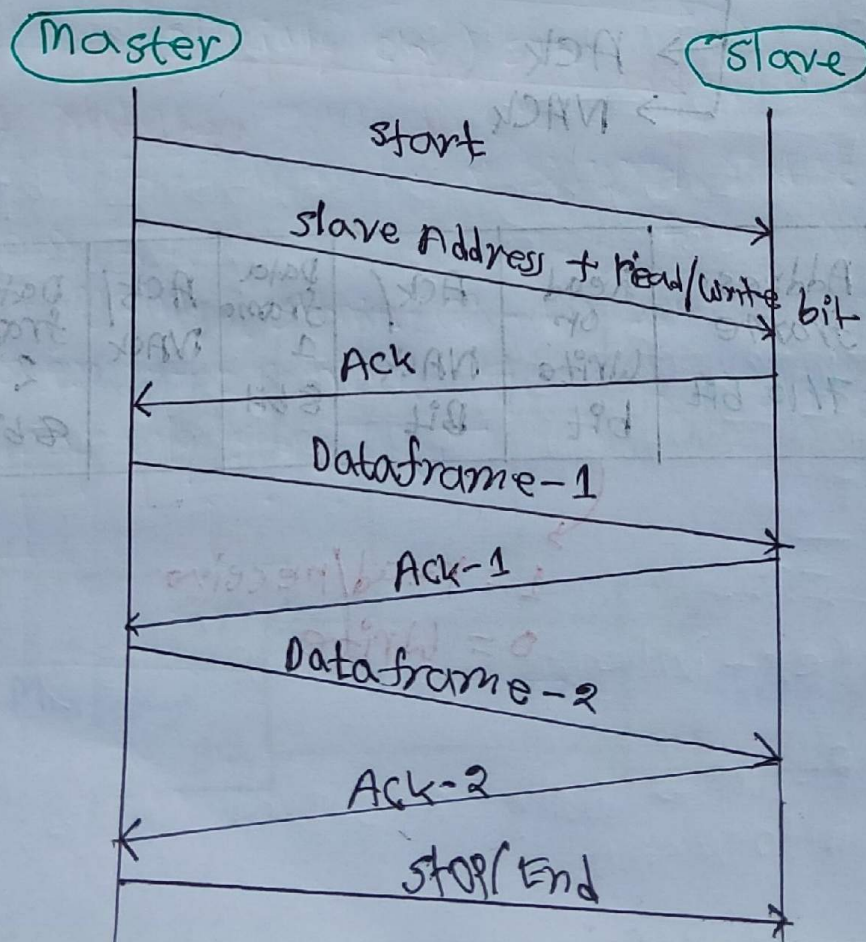
↳ Ack
↳ NACK

start	Address frame 7/10 bit	Read or Write bit	Ack/ NACK Bit	Data frame 1 8 bit	Ack/ NACK	Data frame 2 8 bit	Ack/ NACK	End
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1 = read/receive
0 = Write.

Handshaking in I2C:

- i) Master asserts a 'start' condition on the bus slaves become awake with this
- ii) Master sends the address of the selected slaves along with read/write bit
- iii) Every slave will check if the address is of him. If yes, it will send back an Ack.
- iv) Now the connection is established they can send data frame. Against each data frame, the receiver sends back an 'Ack'



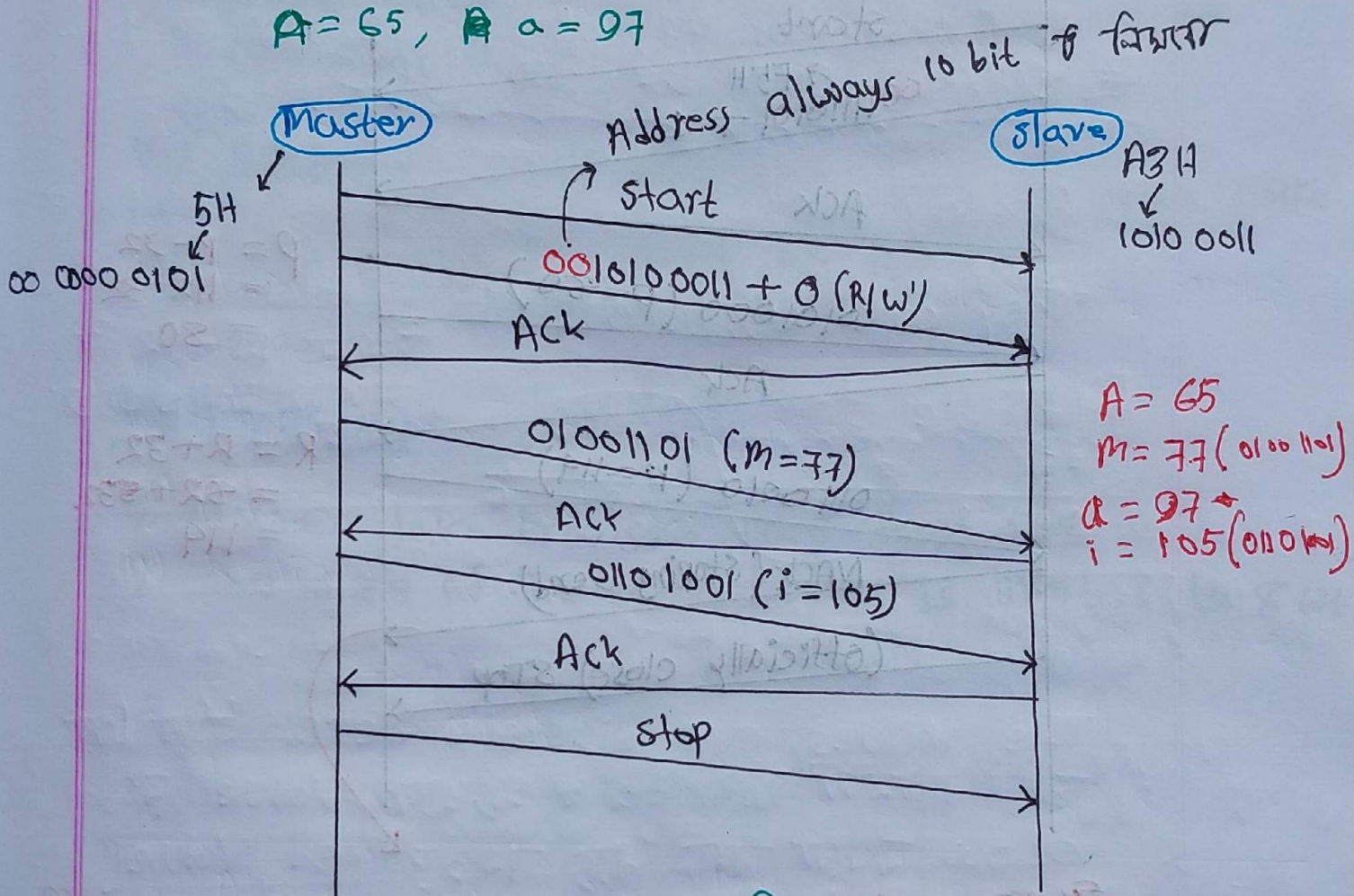
send = write
Receive = Read.

Spring - 25 Q2(b)

master master.

N.B i) First Letter is 1 Byte, 2 Byte is 2 Byte
~~Letter is 1 Byte, 2 Byte is 2 Byte~~

A = 65, a = 97



2 byte master stop signal

N.B

if master is miss the data slave NACK master NACK master
Data master, Repeat the
until Ack master.

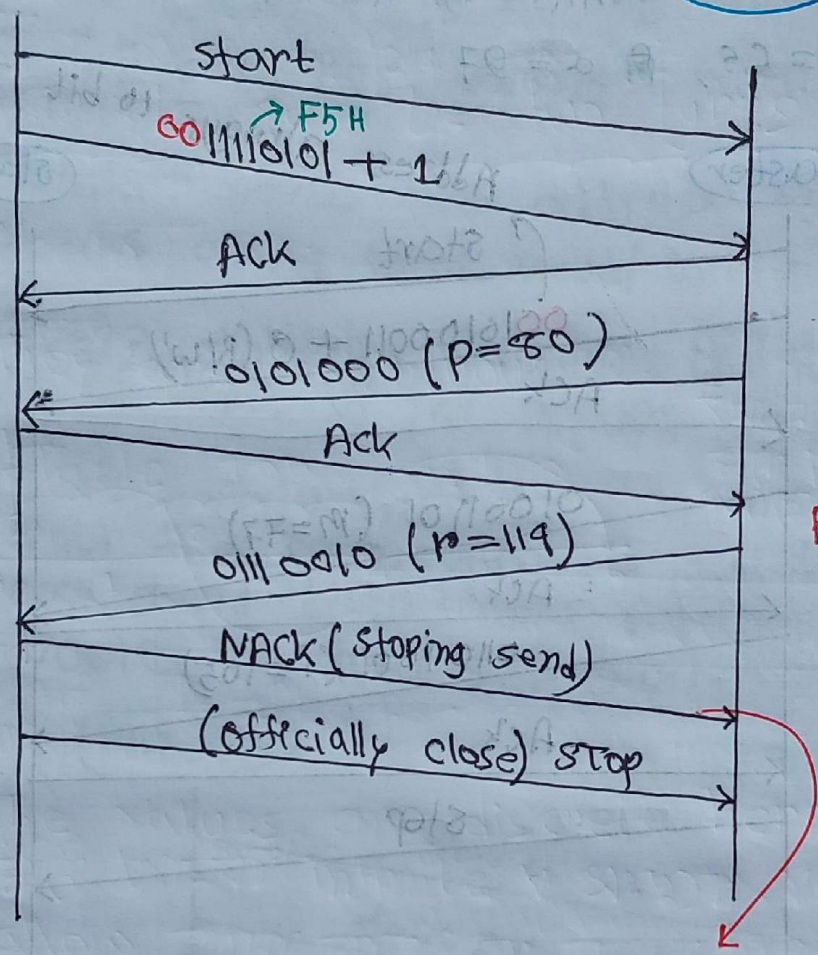
Trick
 $A = 65, a = 97$
 capital to small $\rightarrow +32$
 small to capital $\rightarrow -32$

Summer 2025 Q2(b)

master $\rightarrow$ receive $\rightarrow$ ~~stop~~

Master

Slave



$$P = P - 32$$

$$= 112 - 32$$

$$= 80$$

$$R = R + 32$$

$$= 82 + 32$$

$$= 114$$

Q2(b) mainly slave is direct stop or master master NACK. For master slave to temporarily stop ~~stop~~

Fall 25 Q 1 (b)

master $\rightarrow$ receive করে।

- i) start should be sent from master.
- ii) slave sent 'e'. It should be 'E' = 69
 $69 = 01000101 (E)$
- iii) Before 'stop', master has to send 'NACK'
- iv) stop should be sent by master.

DisAdvantage:

- i) slower data transfer rate than SPI
- ii) The size of the data frame is limited to 8 bit

